Vivek Shukla

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GitHub | LeetCode | GfG | Portfolio | LinkedIn | Medium

Objective

MCA student with hands-on experience building Python-based AI/ML systems, from NLP text classification to LLM agents with tool calling, guardrails, and deterministic business rules. Seeking an AI/ML or Generative AI role building reliable, secure intelligent applications.

Education

Jawaharlal Nehru University, New Delhi 2025 – 2027
Master of Computer Applications (MCA), CGPA: 7.54 / 9

M.L.K.P.G. College, Balrampur Sep 2021 – May 2024
Bachelor of Computer Applications (BCA), CGPA: 8.12 / 10

Technical Skills

- **Languages:** Python, Java, SQL, TypeScript
- **AI / ML:** LangGraph, LLM Agents & Tool Calling, Prompt Engineering, Gemini API, NLP (NLTK), TF-IDF, Prompt-Injection Defense
- **Backend & Data:** FastAPI, REST APIs, SQLAlchemy, SQLite, Server-Sent Events (SSE)
- **Tools & Apps:** Docker, Git, GitHub, Postman, Streamlit, Tkinter, React.js, Linux (command-line basics)
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Experience

MIS Assistant — People Action for National Integration (PANI) *Jul 2025 – Oct 2025*

- Validated and audited project datasets in MS Excel and MIS tools, applying detail-oriented analysis to identify and resolve discrepancies.
- Collaborated with internal teams and external stakeholders to streamline reporting workflows, building customer-facing communication and documentation skills.

Projects

AI Customer Support Agent GitHub
Python, FastAPI, LangGraph, React, TypeScript, SQLite, SQLAlchemy, Docker

Problem: Manual e-commerce refund processing is slow and error-prone. This agent automates customer verification, order lookup, refund-policy evaluation, and escalation, with real-time admin monitoring.

- **Workflow:** 6-tool LangGraph agent for customer verification, order lookup, eligibility checks, refunds, and human escalation.
- **Business rules:** Deterministic 30-day return policy and $500 escalation threshold.
- **Security:** Identity validation, order-ownership checks, prompt-injection detection, and output sanitization.
- **Guardrails:** 2,000-character input limit, 30-message cap, 60K-token budget, 10 requests/minute rate limit.
- **Monitoring:** Real-time admin dashboard using SSE to stream agent reasoning and refund decisions.
- **Test data:** Mock CRM with 15 customers and 20 orders covering approved, denied, and escalated cases.

Spam Detection System (NLP + ML) GitHub
Python, NLTK, TF-IDF, Machine Learning, Streamlit, Tkinter

Problem: Spam messages cause inconvenience, fraud risk, and reduced user trust. This system classifies SMS/email messages as spam or legitimate with accuracy and speed.

- **Preprocessing:** NLTK pipeline with lowercasing, tokenization, stopword and punctuation removal, and stemming.
- **Model:** TF-IDF vectorization with a pre-trained ML model for real-time classification of new messages.
- **Interface:** Streamlit web app and a standalone Tkinter desktop GUI covering the full ML pipeline.

Achievements

- Ranked in the top 30% out of 2,989 participants in the OpenAI Codex Hackathon, organized by NamasteDev.

Certifications

- Machine Learning Summer Training — Analytics Vidhya (Link)
- HackerRank: Software Engineer (Link) and Software Engineer Intern (Link) Certificates